

SIGNAL & DATA ANALYSIS IN NEUROSCIENCE 2020 INFORMATION THEORY

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Outline

- Entropy
- Mutual information

Entropy: definitions

- **Entropy** of a system is the uncertainty about the state of that system. It is the expected number of bits required to fully describe the state of the system.

- Let X be a discrete RV with a given distribution $p(x)$

$$H(X) = -\sum_x p(x) \log_2[p(x)]$$

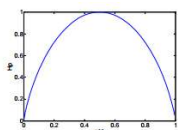
- The units are "bits" and when using $\ln$ the units are "nats"
- Notice : the values of the variable are not important, only its distribution

Example

- Let X be a random variable that get 1 with probability of p and 0 Otherwise

$$H(X) = -[p \log_2(p) + (1-p) \log_2(1-p)]$$

- The entropy is maximum with uniform distribution and minimum when there is only one possible value.



Example

- Suppose we have random variable with uniform probability over 8 values (it is the index of winning horse in a race)

- Q: how many bits are needed to encode this variable ?

- The entropy of this variable:

$$H(X) = -\sum_{i=1}^8 p(i) \log p(i) = -\sum_{i=1}^8 1/8 \log 1/8 = \log 8 = 3 \text{ bits}$$

Non uniform distribution...

- Now suppose the distribution of winning horse is not uniform, but

(1/2, 1/4, 1/8, 1/16, 1/64, 1/64, 1/64, 1/64)

Now the entropy would be:

$$H(X) = -1/2 \log 1/2 - 1/4 \log 1/4 - 1/8 \log 1/8 - 1/16 \log 1/16 - 4 * 1/64 \log 1/64 = 2 \text{ bits}$$

- If we want to send the winning horse index over a channel we can send its binary form, or we can do better, and use this representation to for the horses

0, 10, 110, 1110, 111100, 111101, 111110, 111111

The mean length per message will be then 2 bits

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Definition: Mutual and Conditional Entropy

- Conditional Entropy

$$\begin{aligned} H(Y|X) &= \sum_{x_i \in \mathcal{X}} p(x_i) H(Y|x = x_i) \\ &= \sum_{x_i \in \mathcal{X}} p(x_i) \sum_{y_j \in \mathcal{Y}} -p(y_j|x_i) \log(p(y_j|x_i)) \end{aligned}$$

- The mutual entropy of 2 random variables with mutual distribution $p(x,y)$ is defined as:

$$\begin{aligned} H(X, Y) &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \cdot \log p(x, y) = \\ &= H(X) + H(Y|X) \text{ /chain rule} \end{aligned}$$

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Properties

- $0 \leq H(X) \leq \log |\mathcal{X}|$
- $H(X|Y) \leq H(X)$
- $H(X|Y) \neq H(Y|X)$
- $H(X, Y|Z) = H(X|Z) + H(Y|X, Z)$
- $H(X_1, X_2, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$

$$H(x_1, x_2, \dots, x_n) = \sum_{i=1}^n H(x_i | x_{i-1}, x_{i-2}, \dots, x_1)$$

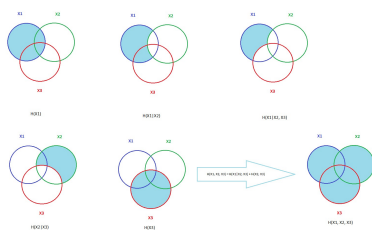
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Joint entropy – 3 Variables

- $H(\mathbf{X}, \mathbf{Y}) = H(\mathbf{Y}|\mathbf{X}) + H(\mathbf{X})$

Example for 3 variables:

- $H(X_1, \mathbf{X_2}, \mathbf{X_3}) = H(X_1|\mathbf{X_2}, \mathbf{X_3}) + H(\mathbf{X_2}, \mathbf{X_3})$,
 $H(\mathbf{X_2}, \mathbf{X_3}) = H(\mathbf{X_2}|\mathbf{X_3}) + H(\mathbf{X_3})$



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Example: Entropy

Let (X, Y) have the following joint distribution:

$Y \backslash X$	1	2	3	4
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{4}$	0	0	0

- Find: $H(X)$, $H(Y)$
- Find: $H(X|Y)$, $H(Y|X)$
- Find: $H(X, Y)$

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Solution

The marginal distribution of X is $(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ and the marginal distribution of Y is $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, and hence $H(X) = 7/4$ bits and $H(Y) = 2$ bits. Also,

$$\begin{aligned}
 H(X|Y) &= \sum_{i=1}^4 p(Y=i) H(X|Y=i) \\
 &= \frac{1}{4} H\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right) + \frac{1}{4} H\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}\right) \\
 &\quad + \frac{1}{4} H\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right) + \frac{1}{4} H(1, 0, 0, 0) \\
 &= \frac{1}{4} \times \frac{7}{4} + \frac{1}{4} \times \frac{7}{4} + \frac{1}{4} \times 2 + \frac{1}{4} \times 0 \\
 &= \frac{11}{8} \text{ bits.}
 \end{aligned}$$

$Y \backslash X$	1	2	3	4
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{4}$	0	0	0

Similarly $H(Y|X) = 13/8$ bits and $H(X, Y) = 27/8$ bits.

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Mutual information

- Definition: $I(X; Y)$ is the relative entropy between the joint distribution and the product distribution $p(X)p(Y)$.

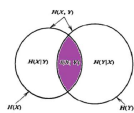
$$\begin{aligned}
 I(X; Y) &= \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \\
 &= H(X) - H(X|Y) = H(Y) - H(Y|X) \\
 &= H(X) + H(Y) - H(X, Y)
 \end{aligned}$$

Properties:

$$I(X; X) = H(X) - H(X|X) = H(X)$$

$$0 \leq I(X; Y) \leq \min(H(X), H(Y))$$

Data processing inequality: if $X \rightarrow Y \rightarrow Z$ $I(X; Y) \geq I(X; Z)$



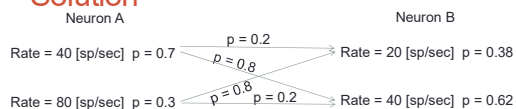
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Example: mutual information

- Neuron A may be recorded at two rates: 40spikes/s ($p=0.7$) or 80spikes/s ($p=0.3$). Neuron B fires at two rates as well: 20spikes/s or 40spikes/s.
- When A fires at 40spikes/s the probability of B to fire 40spikes/s is 0.8.
- When A fires at 80spikes/s the probability of B to fire 20spikes/s is 0.8.
- Calculate the entropy of A, B
- Calculate the conditional entropy of B given A .
- Calculate the mutual information.

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Solution



- $H(A) = H(0.7, 0.3) = 0.88$
- $H(B) = H(0.38, 0.62) = 0.96$
- $H(B|A) = \sum_A P(A) * H(B|A) = 0.7_{A=40} * H(0.2, 0.8) + 0.3_{A=80} * H(0.2, 0.8) = H(0.2, 0.8) = 0.72$
- $I(B;A) = H(B) - H(B|A) = 0.24$
- Method II:
 $H(B,A) = H(0.14, 0.56, 0.24, 0.06) = 1.60$
 $I(B;A) = H(A) + H(B) - H(B,A) = 0.24$

Rate A	Rate B	P
40	20	.14
40	40	.56
80	20	.24
80	40	.06

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Example:

- The following table describes the probability of a patient coming into the clinic to be suffering from either Tourette syndrome (TS) or Obsessive Compulsive Disorder (OCD):

TS	OCD	P
-	-	0.9
-	+	0.06
+	-	0.02
+	+	0.02

- Calculate the entropy of TS and OCD and their joint entropy, explain the result.
- Calculate the mutual information between the two disorders, explain the result

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Solution

- $P(TS = +) = 0.04$, $H(TS) = H(0.04, 0.96) = 0.24$
- $P(OCD = +) = 0.08$, $H(OCD) = H(0.08, 0.92) = 0.40$
- $H(TS, OCD) = -\sum P(TS_i, OCD_j) \log(P(TS_i, OCD_j))$
 $= -.9 * \log .9 - .06 * \log .06 - .02 * \log .02 - .02 * \log .02 = 0.60$
- Method I: $I(TS; OCD) = H(TS) + H(OCD) - H(TS, OCD) = 0.04$
- Method II: $I(TS; OCD) = H(TS) - H(TS|OCD)$
 $H(TS|OCD) = P(OCD = +) H(.02/.08, .06/.08)$
 $+ P(OCD = -) H(.90/.92, .02/.92)$
 $= .06 + .14 = 0.2$
 $I(TS; OCD) = H(TS) - H(TS|OCD) = 0.24 - 0.2 = 0.04$
- There is almost no mutual information between TS & OCD.

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Example exam 2006 moed B

- The following table describes the probability a patient Testis Nervosis suffers from two symptoms (S1 & S2).

S1	S2	P
-	-	0.4
-	+	0.3
+	-	0.1
+	+	0.2

- Two random variables are constructed:
 X – has 4 possible values based on the S1 & S2 symptom occurrence.
 Y – has 3 possible values based on the number of symptoms occurring (0, 1, 2)
- Calculate the entropy of X and the entropy of Y, explain the results.
- Calculate the mutual information between the variables, explain the result.

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Solution

- $H(X) = H(.4, .3, .2, .1)$
 $= 1.84$
- $H(Y) = H(.4, .3 + .1, .2)$
 $= 1.5$
 (3 states max 1.65)

S1	S2	P	X	Y
-	-	0.4	1	0
-	+	0.3	2	1
+	-	0.1	3	1
+	+	0.2	4	2

- $H(X; Y) = H(X)$
 $I(X; Y) = H(X) + H(Y) - H(X; Y) = H(Y) = 1.5$
- $H(Y|X) = 0$
 $I(X; Y) = H(Y) - H(Y|X) = H(Y) = 1.5$
 Given X we know Y perfectly.
